SON CAO

Computer Science Student

CONTACT

+31 616411761

chson230905@gmail.com

hoagson2309

Son Cao

230 Zweringweg, 7545DA, Enschede

EDUCATION

Bachelor of Computer Science

Saxion University of Applied Science

2024 - 2028

SKILLS

- C / C++
- Python
- HTML / CSS
- C# / JavaScript
- Git
- Database / SQL
- Machine Learning
- Deep Learning
- Embedded systems (Rasberry Pi Pico, ESP32)

LANGUAGE

English

Vietnamese

PROFILE

I'm a Computer Science student at Saxion University of Applied Sciences with a strong interest in Artificial Intelligence and machine learning. I enjoy learning by building and working on hands-on projects involving data, algorithms, and software development.

I'm proactive, adaptable, and always looking for ways to improve my skills. Currently, I'm seeking internships or part-time opportunities in AI, data or software development where I can gain practical experience and contribute to real-world projects.

PROJECTS

Real-Time Hand Gesture Recognition (CNN + Temporal Convolutions)

Python | TensorFlow/Keras | OpenCV

- Developed a real-time gesture recognition system using CNN for spatial feature extraction and Conv1D (TCN) for temporal sequence modeling
- Built a video preprocessing pipeline (frame extraction, normalization, sliding window sequence generation) using OpenCV
- Trained on a custom dataset (thumbs up, thumbs down, stop) and achieved high validation accuracy
- Deployed real-time inference via webcam with low-latency predictions on a standard laptop
- Applied spatio-temporal learning to effectively model motion in video-based gesture classification

NVIDIA (NVDA) Stock Price Prediction

Python | scikit-learn | Pandas | Matplotlib | Alpaca API

- Developed a Random Forest model to predict daily returns of NVDA stock using historical data (2022–2026)
- Engineered features using technical indicators (SMA, RSI, volatility) to capture market trends
- Achieved ~1.7% MAE on daily returns, consistent with expected market noise
- Applied a time-aware train/test split to preserve chronological data integrity
- Visualized performance through actual vs predicted returns and cumulative return analysis
- Analyzed model limitations, demonstrating challenges of predicting volatile financial time-series data and the impact of compounding errors

Autonomous Maze Navigation Robot (Raspberry Pi Pico)

C++ | Raspberry Pi Pico | HC-SR04 | ITR20001 | PCA9685 | TB6612FNG | Embedded Systems | DFS | A* | CMake

- Built an embedded robotics system for autonomous maze exploration, real-time mapping, and navigation on Raspberry Pi Pico
- Designed a 20×20 grid-based mapping system using ultrasonic and infrared sensors for wall detection, environment perception and obstacle detection.
- Implemented DFS-based exploration with backtracking to traverse unknown mazes efficiently
- Developed A* pathfinding algorithm to compute the optimal return path after goal detection
- Engineered motor control using TB6612FNG and PCA9685 for precise movement, alignment, and obstacle handling